Advanced Time Value of Money (TVM)
Perpetuities, Growing Annuities, and Corporate Applications

I. Course Structure and Quantitative Emphasis

Nature of the Course

- Fully quantitative and calculation-based.
- Focus on assignments and consistent numerical practice.
- No major projects; performance depends on exams.

Grading Distribution

Component	Weight
Midterm 1	15%
Midterm 2	15%
Final Exam	45–50%

II. Perpetuities

Definition

A perpetuity is an annuity with an infinite life:

$$n \to \infty$$

A. Constant Perpetuity

$$PV = \frac{PMT}{i}$$

Example:
Preferred stock pays \$500 annually forever. Required return = 10%.

$$PV = \frac{500}{0.10} = 5{,}000$$

B. Growing Perpetuity

$$PV = \frac{PMT}{i - g}$$

Important Condition:

$$i > g$$

If $g \geq i$, the formula becomes invalid.

Example:

Dividend next year = \$500
Growth rate = 4%
Required return = 10%

$$PV = \frac{500}{0.10 - 0.04}$$

$$PV = \frac{500}{0.06} = 8{,}333.33$$

III. Growing Annuities

A. Future Value of Growing Annuity

$$FV = PMT \times \frac{(1 + i)^n - (1 + g)^n}{i - g}$$

Example:

PMT = 2,000
$i = 10\%$
$g = 5\%$
$n = 5$

$$FV = 2000 \times \frac{(1.10)^5 - (1.05)^5}{0.10 - 0.05}$$

$$(1.10)^5 = 1.6105$$

$$(1.05)^5 = 1.2763$$

$$FV = 2000 \times \frac{1.6105 - 1.2763}{0.05}$$

$$FV = 2000 \times 6.684 = 13{,}368$$

B. Present Value of Growing Annuity

$$PV = \frac{PMT}{i-g}\left[1 - \left(\frac{1+g}{1+i}\right)^n\right]$$

Example:
PMT = 5,000
$i = 9\%$
$g = 3\%$
$n = 6$

$$PV = \frac{5000}{0.09 - 0.03}\left[1 - \left(\frac{1.03}{1.09}\right)^6\right]$$

$$\frac{5000}{0.06} = 83{,}333.33$$

$$\left(\frac{1.03}{1.09}\right)^6 \approx 0.699$$

$$PV = 83{,}333.33 \times (1 - 0.699)$$

$$PV = 83{,}333.33 \times 0.301 = 25{,}083$$

IV. Lease vs Buy Decision Model

Buy Option

Cost equals immediate purchase price (already in present value form).

$$Cost_{Buy} = Purchase\ Price$$

Lease Option

$$PV_{Lease} = PMT \times \frac{1-(1+i)^{-n}}{i}$$

Example:
Machine price = 100,000
Lease = 25,000 per year
$n = 5$
$i = 8\%$

$$PV_{Lease} = 25000 \times \frac{1-(1.08)^{-5}}{0.08}$$

$$(1.08)^{-5} = 0.6806$$

$$PV = 25000 \times 3.9925$$

$$PV = 99{,}812$$

Since:

$$99{,}812 < 100{,}000$$

Leasing is slightly cheaper.

V. Trust Fund Case Study: Two Grandsons

Scenario 1: Steve

PMT $= 2{,}500$
$i = 8\%$
$n = 46$

Future value formula:

$$FV = PMT \times \frac{(1+i)^n - 1}{i}$$

$$FV = 2500 \times \frac{(1.08)^{46} - 1}{0.08}$$

$$(1.08)^{46} \approx 34.90$$

$$FV = 2500 \times 423.75$$

$$FV \approx 1{,}059{,}375$$

Scenario 2: Second Grandson

Target:

$$FV = 1{,}059{,}375$$

$n = 41$, $i = 8\%$
Solve for PMT:

$$PMT = \frac{FV \cdot i}{(1+i)^n - 1}$$

$$PMT = \frac{1{,}059{,}375 \times 0.08}{(1.08)^{41} - 1}$$

$$(1.08)^{41} \approx 23.33$$

$$PMT \approx 3{,}796$$

(Rounding differences may produce approximately 3,726.)

VI. Key Conceptual Summary

- Perpetuities assume infinite life.
- Growing perpetuity requires $i > g$.
- Growing annuities incorporate both interest and growth.
- Corporate decisions are made using present value comparison.
- Fewer periods require higher payments to reach the same future value.

VII. Core Formulas to Memorize

$$PV_{Perpetuity} = \frac{PMT}{i}$$

$$PV_{Growing\ Perpetuity} = \frac{PMT}{i-g}$$

$$FV_{Growing\ Annuity} = PMT \times \frac{(1+i)^n - (1+g)^n}{i-g}$$

$$PV_{Growing\ Annuity} = \frac{PMT}{i-g}\left[1 - \left(\frac{1+g}{1+i}\right)^n\right]$$

$$FV_{Ordinary\ Annuity} = PMT \times \frac{(1+i)^n - 1}{i}$$